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Thesis Proposal

Design of a Scalable Hybrid I/O CGRA with Multicast Routing for Efficient Edge SNN Inference

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Abstract

This thesis presents the architectural definition and implementation of a Coarse-Grained Reconfigurable Architecture (CGRA) optimized for Spiking Neural Network (SNN) inference. Modern event-driven workloads suffer from irregular data access and significant I/O bottlenecks. To address these challenges, the proposed accelerator features a novel "Hybrid I/O" architecture that decouples high-bandwidth input streaming (DMA) from low-latency control and status output (APB), effectively minimizing bus contention and reducing synchronization overhead.

A key innovation of the design is the integration of a parameterizable mesh of Processing Elements (PEs) interconnected by a dynamic multicast-capable routing fabric. This fabric exploits the high fan-out nature of SNNs by replicating spike packets to multiple destination PEs in a single cycle, significantly reducing latency compared to traditional point-to-point networks. Each PE includes a hardware-accelerated Leaky Integrate-and-Fire (LIF) unit to compute neuron dynamics with high energy efficiency.

The system is implemented on the Xilinx Zynq UltraScale+ platform using a hardware-software co-verification strategy. Key technical contributions include a double-buffered configuration memory for zero-latency reconfiguration and a deterministic scheduling mechanism for sparse dataflow execution. Experimental results confirm a maximum operating frequency of 213.72 MHz and 100% pass rate across rigorous regression suites, establishing a scalable and robust platform for edge-based neuromorphic computing.

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List of Abbreviations

ALU	Arithmetic Logic Unit
AMBA	Advanced Microcontroller Bus Architecture
ANN	Artificial Neural Network
APB	Advanced Peripheral Bus
ASIC	Application-Specific Integrated Circuit
AXI	Advanced eXtensible Interface
BPTT	Backpropagation Through Time
CGRA	Coarse-Grained Reconfigurable Architecture
CPU	Central Processing Unit
CSC	Compressed Sparse Column
CSR	Control and Status Register
DDR	Double Data Rate (Memory)
DMA	Direct Memory Access
DRAM	Dynamic Random Access Memory
DVS	Dynamic Vision Sensor
FIFO	First-In, First-Out
FPGA	Field-Programmable Gate Array
FSM	Finite State Machine
GPU	Graphics Processing Unit
IP	Intellectual Property
LIF	Leaky Integrate-and-Fire
LUT	Lookup Table
MAC	Multiply-Accumulate
NoC	Network on Chip

PE	Processing Element
RAM	Random Access Memory
RF	Register File
SIMD	Single Instruction, Multiple Data
SNN	Spiking Neural Network
SoC	System on Chip
SPM	Scratchpad Memory
SRAM	Static Random Access Memory
STDP	Spike-Timing-Dependent Plasticity
VLIW	Very Long Instruction Word

1 Introduction

1.1 Overview of CGRA and SNN Inference

Spiking Neural Networks (SNNs) represent the third generation of neural computation [1], providing energy-efficient execution through event-driven mechanisms. However, deploying SNNs on traditional hardware (CPUs/GPUs) is inefficient due to sparse, irregular connectivity. SNNs operate on discrete spike events, causing SIMD utilization to drop sharply in von Neumann processors.

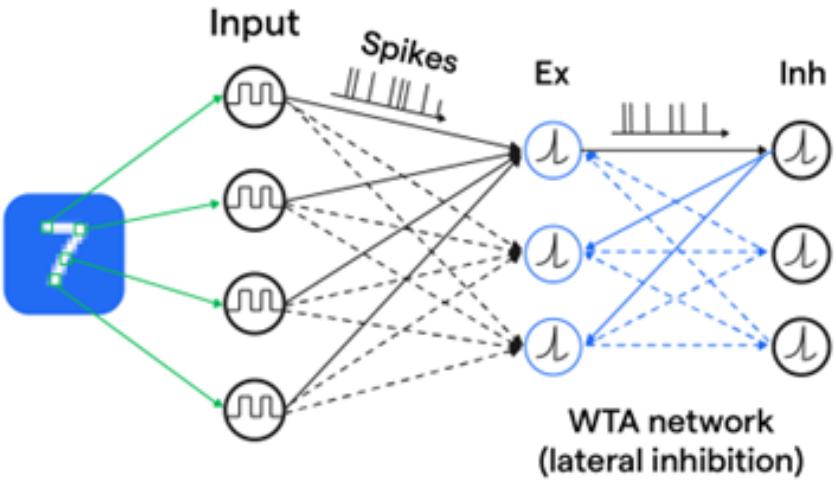

Figure 1: Biological vs. Artificial Spiking Neuron Models

To address these limitations, the proposed CGRA employs a spatially programmable 4×4 mesh of Processing Elements (PEs). Each tile combines a fixed-point Multiply-Accumulate (MAC) unit with a hardwareimplemented Leaky IntegrateandFire (LIF) neuron update engine. This unified execution model allows the array to compute sparse matrix–vector (matvec) operations and neuron dynamics in a tightly coupled, dataflow-driven manner.

1.2 System Context and Execution Flow

The CGRA is designed as a tightly integrated hardware accelerator under full control of an ARM Cortex-A53 subsystem. The typical execution flow consists of:

1. **Configuration Stage:** The host processor writes bitstream (context words) into the CGRA configuration memory via the CSR interface. These bitstream encode the spatial and temporal behavior of the 4×4 PE mesh.

2. **Data Preparation:** SNN weight matrices are stored in Compressed Sparse Column (CSC) format in DDR memory. Input spike vectors and neuron states are prepared and aligned for DMA-based transfer.

3. **Execution Phase:** The CGRA performs sparse matvec operations and LIF neuron updates in each PE according to the dynamical equation: $v[t+1] = \beta \cdot v[t] + I[t] - \lambda$.

4. **Result Collection:** Updated membrane potentials and spike outputs are written back to DDR for use by the next simulation timestep or software post-processing.

1.3 Target Workload and Problem Scope

The CGRA accelerator is optimized for sparse SNN layer updates, specifically weight matrices $W \in \mathbb{R}^{128\times128}$ stored in CSC format. This representation allows the accelerator to process only active presynaptic spikes by scanning the corresponding nonzero column entries, thus avoiding dense computation. The primary bottlenecks addressed are:

- **Routing Congestion:** fan-out dominant distributions causing "Routing Black Holes."
- **Data Starvation:** irregular memory access patterns leading to PE idle cycles.
- **Precision Inefficiency:** inefficient use of bit-width in standard generic ALUs, requiring specialized 40-bit saturating logic for high-fidelity LIF updates.

The system architecture and DMA engine are designed according to the AMBA AXI Standard [2], ensuring interoperability with the Zynq UltraScale+ processing system [3].

2 Literature Review and Background

2.1 Overview of Coarse-Grained Reconfigurable Architectures

Coarse-Grained Reconfigurable Architectures (CGRAs) represent a class of programmable hardware structures positioned between traditional fine-grained FPGAs and fixed-function ASICs. Well-known examples include ADRES [4] and the more recent Plasticine architecture [5]. Extensive surveys on CGRA performance and architectural diversity can be found in [6] and [7].

CGRAs mitigate issue of routing congestion and configuration overhead by operating at the arithmetic-operator granularity. A typical CGRA execution flow involves configuration mapping, spatial/temporal scheduling, and data routing across the mesh.

2.2 Spiking Neural Networks (SNNs)

SNNs represent the third generation of neural network models [1], in which information is encoded and processed using discrete spike events rather than continuous-valued activations. This temporal and event-driven formulation, as detailed in [8], allows SNNs to more closely resemble biological neural systems. Modern deep learning applications leveraging these models are extensively discussed in [9] and [10], while training methodologies such as surrogate gradient learning [11] have bridged the gap between SNNs and traditional deep learning.

2.2.1 Leaky Integrate-and-Fire (LIF) Neuron Model

Among many neuron models, the LIF model strikes a balance between biological plausibility and implementation simplicity. It captures the essential features of integration, leakage, and threshold-based spiking. In discrete time, the LIF neuron is modeled as:

$$v[t+1] = \beta \cdot v[t] + I[t] - \lambda \tag{1}$$

where $v[t]$ is the membrane potential, $\beta \in (0, 1)$ is the leak factor, $I[t]$ is the sum of weighted spikes, and λ is a constant modeling tonic leak.

2.2.2 Information Encoding Schemes

There are two primary encoding paradigms supported by this architecture:

- **Rate Coding:** Information is conveyed by the firing rate of the neurons.
- **Temporal Coding:** Information is encoded in the precise timing of individual spikes (Time-to-First-Spike).

2.3 Computational Advantages of CGRA for SNNs

The theoretical motivation for designing a specific CGRA for SNNs lies in:

- **Event-Driven Processing:** Computation is only triggered when an input spike arrives, saving dynamic power.
- **Sparsity Exploitation:** By using CSC format and multicast routing, the architecture avoids fetching "zero" values.
- **Locality of Reference:** Keeping the membrane potential state $v[t]$ in local PE scratchpads reduces data movement energy.

The SNN implementation and dataset verification also leverage standard benchmarks like MNIST [12] and modern software frameworks such as snnTorch [13].

3 Project Objectives

The primary objective of this project is to design and implement a CGRA-based accelerator IP featuring an APB control interface and an AXI4 high-bandwidth DMA engine.

3.1 Design Goals

- **High-Dynamic-Range Computation:** Development of a 4×4 PE mesh incorporating 40-bit saturating arithmetic.
- **Neuromorphic Acceleration:** Integration of dedicated hardware LIF units within each PE for efficient potential evolution and spike generation.
- **Low-Latency Context Switching:** Implementation of a "hot-reloadable" configuration mechanism using double-buffered context memory.
- **Scalable Sparse Processing:** Support for Compressed Sparse Column (CSC) data structures to optimize for connectivity sparsity (<1% activity).

3.2 Evaluation Criteria

The success of the project will be measured by:

1. Correct functional behavior as verified by a 6,010-case randomized regression suite.
2. Protocol compliance with AMBA AXI4 and APB standards as validated by assertions.
3. Timing closure at >200 MHz on the Zynq UltraScale+ platform.
4. Demonstration of end-to-end inference for digit classification tasks (MNIST).

4 Proposed System Architecture

The system architecture is partitioned into four major subsystems designed to maximize data throughput and minimize stall cycles.

4.1 SoC System Integration: The Hybrid I/O Architecture

The CGRA IP is integrated into the Programmable Logic of the Zynq UltraScale+ platform. To overcome the common "bottleneck black hole" found in standard accelerator designs, this project proposes a **Hybrid I/O architecture** that decouples data and control paths:

- **AXI4 Data Plane (DMA):** Facilitates high-bandwidth, burst-oriented communication with DDR4 DRAM for weight and spike streaming.
- **APB Control Plane (CSR):** A dedicated low-latency bus for register-mapped configuration, status monitoring, and interrupt handling, ensuring control transactions never stall the data pipeline.

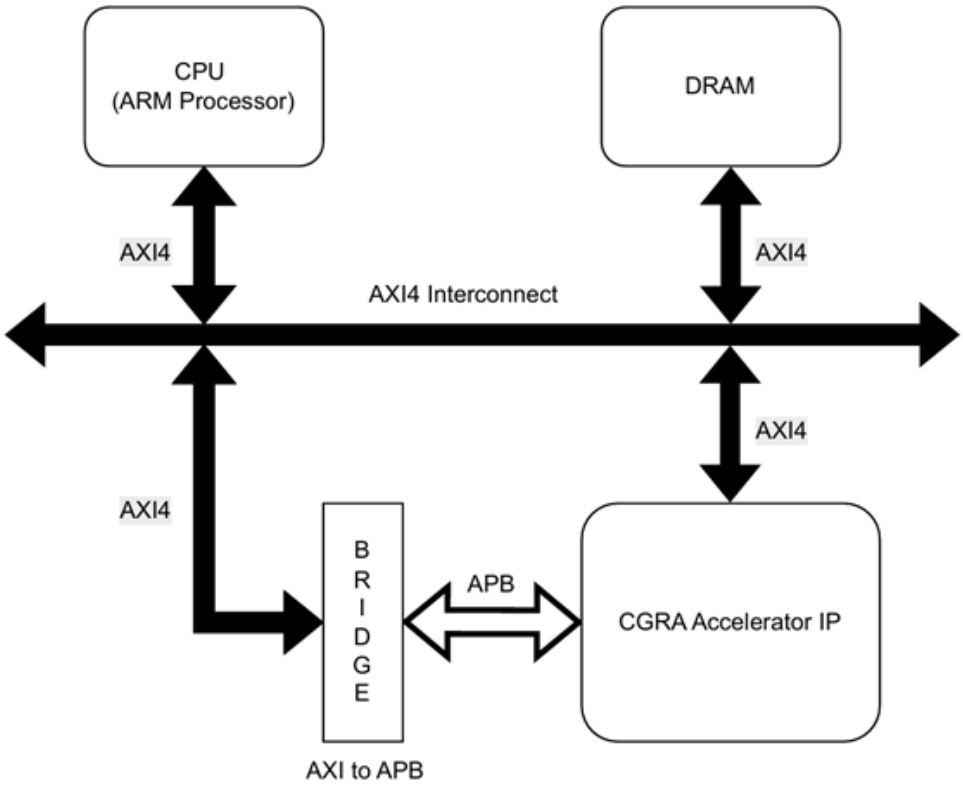

Figure 2: Hybrid I/O Architecture: Decoupled Control (APB) and Data (AXI4) Planes

4.2 Accelerator Subsystems

4.2.1 Global Control and CSR

This module decodes commands from the host via the APB slave interface. It manages interrupt generation, status monitoring, and the accelerator's lifecycle.

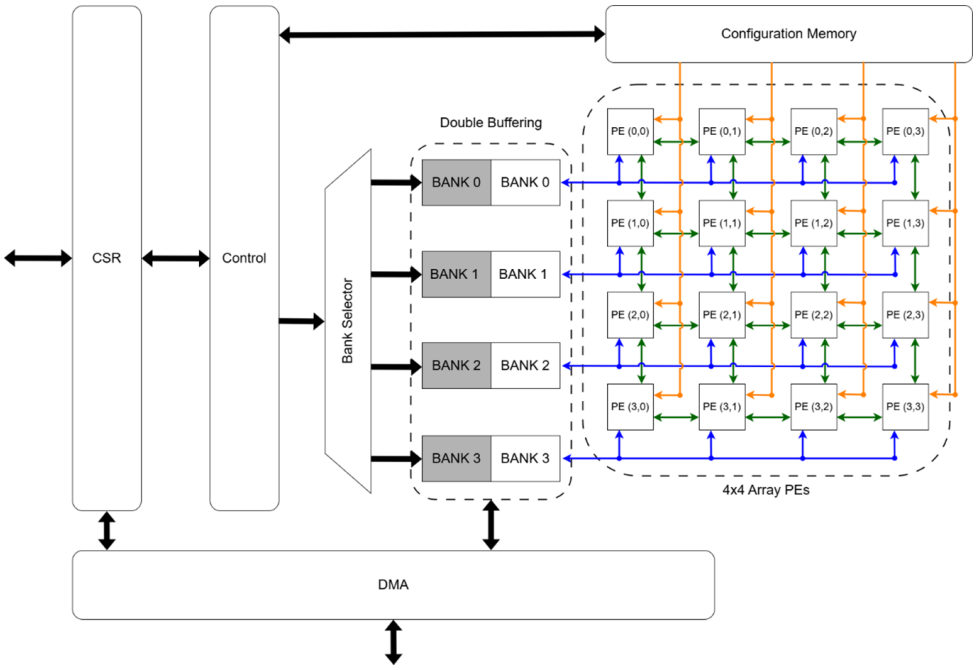

Figure 3: High-Level Block Diagram of the CGRA Accelerator IP Subsystems

4.2.2 Double-Buffered Configuration Memory

To support rapid reconfiguration required by multi-layer networks, the configuration memory employs a double-buffering scheme. This allows the DMA to pre-fetch the bitstream for the next layer (the "back" buffer) while the current layer is executing from the "front" buffer.

4.2.3 Bi-Directional AXI4 DMA Engine

The DMA unit operates as an AXI4 Master with **backpressure-aware flow control**. Unlike traditional input-only designs, this engine features a dedicated Write Engine to offload intermediate neuron states back to DDR. This enables the execution of multi-layer SNNs that exceed on-chip memory capacity without CPU-side data copying.

4.2.4 Row-Banked Tile Memory

The global tile memory is physically partitioned into four independent SRAM banks. Each bank is dedicated to a specific row of the 4×4 PE array, allowing all four rows to fetch operands simultaneously without contention.

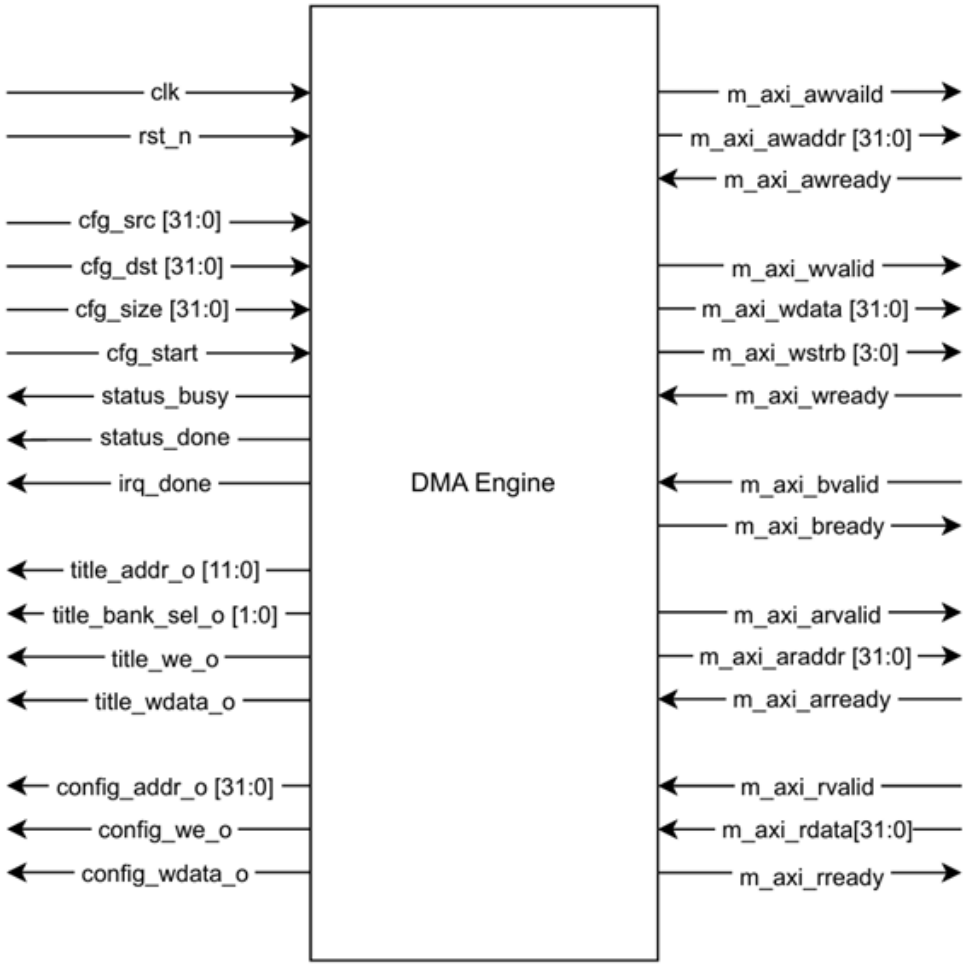

Figure 4: Bi-Directional AXI4 DMA Engine Architecture

4.3 Processing Element Tile Design

Each PE tile integrates a compute datapath, local scratchpad memory, and a **Multicast-Enabled Network-on-Chip (NoC)** router. The PE datapath consists of:

- **Decode:** Configuration frame sets opcode and route mask.
- **Operand Select:** Dual 4-to-1 muxes for source operand selection (RF, Neighbors, SPM, Immediate).
- **ALU:** 40-bit saturating MAC unit and 32-bit ALU.
- **LIF Unit:** Discretized neuron update logic.
- **Router (Multicast-Enabled NoC):** A circuit-switched mesh router. It implements a **dynamic bitmask-based multicast algorithm**, allowing a single spike packet to be replicated to any combination of North, East, South, or West neighbors in a single cycle. This is critical for SNNs with high synaptic fan-out.

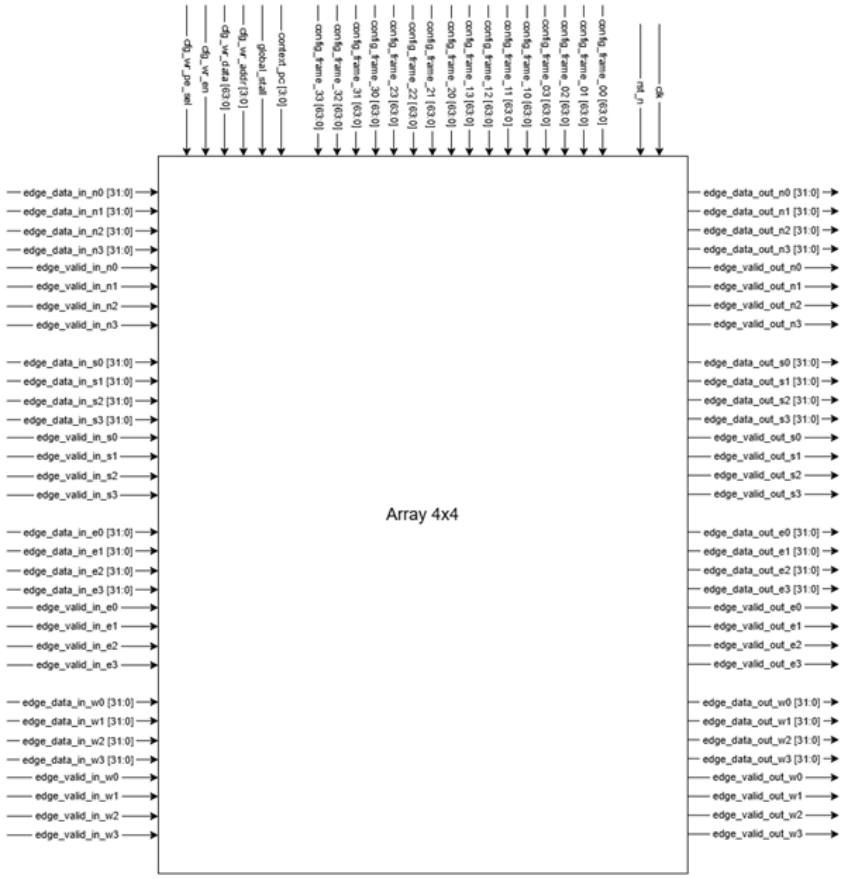

Figure 5: 4x4 PE Array Mesh Topology

4.4 Parametric Design Generation and Scalability

To ensure the proposed architecture is not limited to a fixed 4×4 configuration, the project includes the development of a **parametric RTL generation framework**. Built using SystemVerilog parameters and Python-based templating, this framework allows for design space exploration by automatically generating NxM arrays (e.g., 8×8 or 16×16) with correctly sized NoC routing tables and DMA stride logic.

4.5 Instruction Set Architecture (ISA)

The CGRA operates on a 64-bit Very Long Instruction Word (VLIW) architecture. Each Processing Element (PE) executes one instruction per clock cycle, defined by a 64-bit configuration frame loaded into its local context memory. The ISA supports 19 distinct operations ranging from standard arithmetic to specialized neuromorphic functions.

4.6 Configuration Frame Format

The 64-bit configuration word controls the PE's datapath, including operand selection, ALU operation, destination routing, and immediate value generation. The bit mapping is detailed in Table 1.

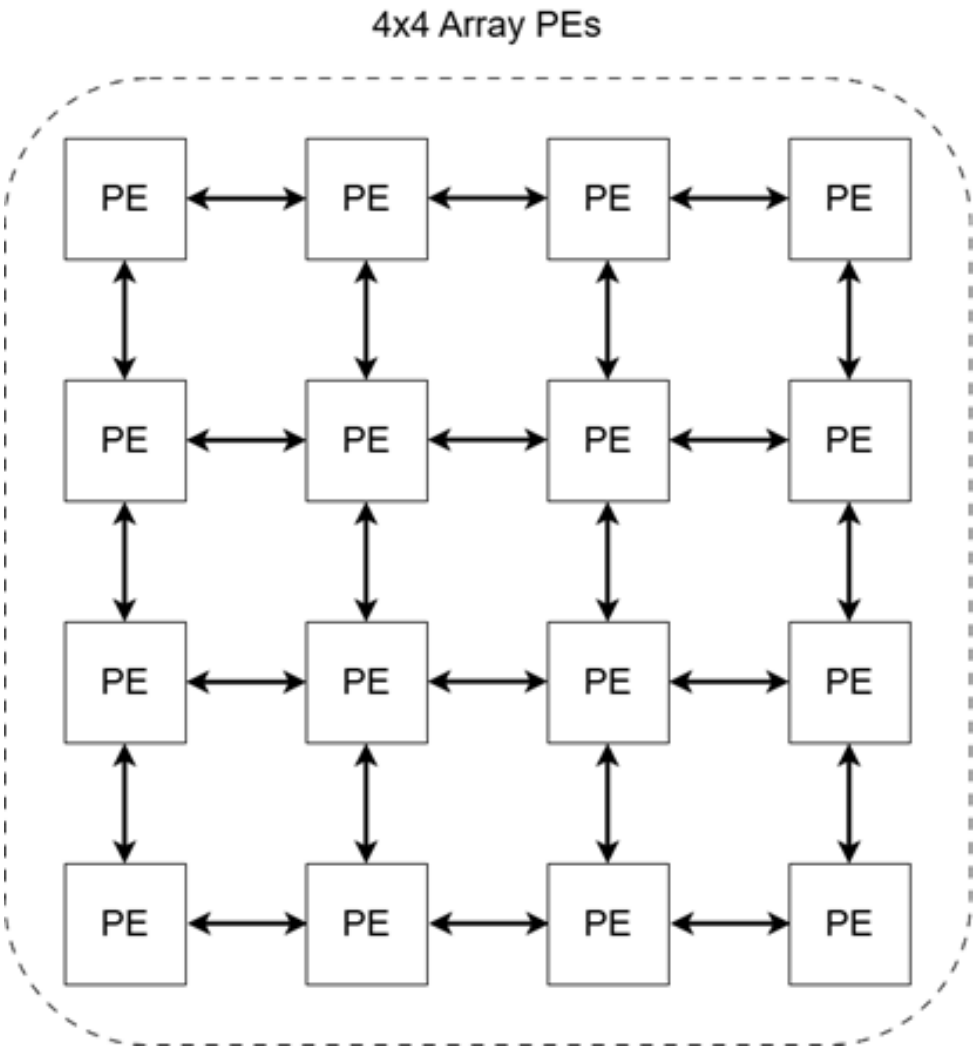

Figure 6: Multicast NoC Routing Patterns in the 4x4 Mesh

Table 1: Configuration Frame Field Descriptions

Field	**Bits**	**Width**	**Description**
opcode	[5:0]	6	ALU Operation Code
src0_sel	[9:6]	4	Source Select for Operand A (RF, Neighbors, SPM, Imm)
src1_sel	[13:10]	4	Source Select for Operand B
dst_sel	[17:14]	4	Write-back address for Local Register File (RF)
route_mask	[21:18]	4	Multicast Output Mask (N, E, S, W)
pred_en	[22]	1	Enable Predicated Execution
pred_inv	[23]	1	Invert Predicate Logic
imm	[39:24]	16	Signed 16-bit Immediate Value
extended	[63:40]	24	Reserved for future extensions

4.7 Opcode Summary

The ALU supports 19 operations categorized into arithmetic, logic, flow control, and neuromorphic updates. All operations execute in a single cycle.

Table 2: ALU Opcode Summary: Arithmetic and Logical Operations

Op	Mnemonic	Description	Op	Mnemonic	Description
0	NOP	No Operation	10	CMP_GT	$(A > B)\ ?\ 1 : 0$
1	ADD	A + B (Saturating)	11	CMP_LT	$(A < B)\ ?\ 1 : 0$
2	SUB	A - B (Saturating)	12	CMP_EQ	$(A == B)\ ?\ 1 : 0$
3	MUL	$A \times B$ (32-bit)	13	LOAD_SPM	$RF \leftarrow SPM[A]$
4	MAC	Acc += $A \times B$ (40-bit)	14	STORE_SPM	$SPM[A] \leftarrow B$
5	AND	A & B (Bitwise AND)	15	ACC_CLR	Acc = 0
6	OR	A | B (Bitwise OR)	16	PASS0	Output = A
7	XOR	$A \oplus B$ (Bitwise XOR)	17	PASS1	Output = B
8	SHL	$A \ll B[4:0]$ (LSL)	18	LIF	Spike = LIF(v, i)
9	SHR	$A \gg B[4:0]$ (ASR)			

4.8 Hardware Signal Interface and Port Definitions

The hardware implementation of the Scalable Hybrid I/O CGRA adheres to industry-standard communication protocols to ensure seamless integration into modern SoC ecosystems. This section defines the signal-level requirements for the four primary RTL modules: the Top-Level Wrapper, the Bi-Directional DMA Engine, the Processing Element (PE), and the Row-Banked Tile Memory.

To maintain high architectural reliability and timing closure, the design follows three fundamental signaling principles:

1. **Protocol Standardization:** All control transactions use the **AMBA APB 3.0** protocol for low-power, non-pipelined configuration access. High-bandwidth data movement is governed by the **AMBA AXI4** (Full) specification, supporting burst transactions and independent read/write channels.

2. **Handshake-Driven Data Flow:** All internal data paths and inter-PE NoC links utilize a **Ready/Valid (Backpressure-Aware)** handshake mechanism. This ensures zero data loss during congestion by stalling upstream producers when downstream consumers are busy.

3. **Unified Synchronous Domain:** All modules operate within a single global clock domain with an active-low synchronous reset (`rst_n`), simplifying the integration into FPGA synthesis flows and preventing Meta-Stability issues across the PE mesh.

The following subsections detail the port definitions for each module, where signal widths marked as "Var" are parameterized to support architectural scaling from 4×4 up to 16×16 arrays.

4.9 Top-Level Module (cgra_top.sv)

The top-level module serves as the primary system wrapper, encapsulating the 4×4 PE array, the bi-directional DMA engine, and the APB-based control unit. It manages the global clock and reset distribution while providing the physical interfaces for SoC-level integration.

The wrapper includes a unified address decoder that maps the 32-bit APB address space to the internal Control Status Registers (CSRs), allowing the host CPU to monitor hardware status, configure DMA descriptors, and trigger execution cycles. Additionally, the top-level unit handles interrupt aggregation, combining DMA-completion and Execution-done signals into a single level-triggered IRQ line for the ARM processor.

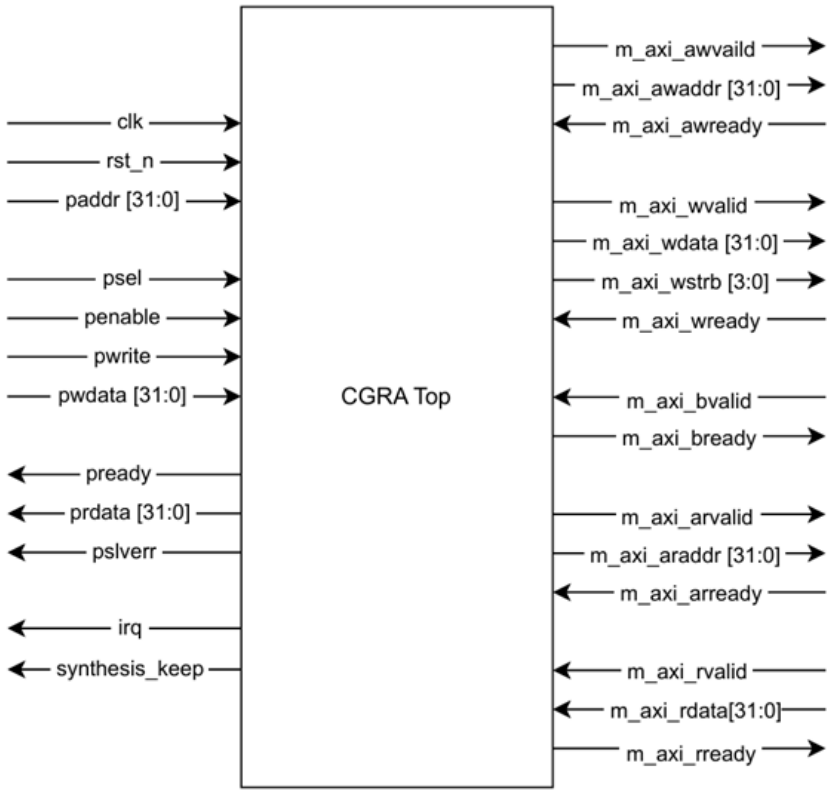

Figure 7: Top-Level RTL Module Hierarchy

Silicon Integration Considerations: The top-level port list is kept purposely minimal to reduce the complexity of the AXI-to-NoC bridge. By parameterizing the address and data widths, the wrapper can be easily adapted to different SoC interconnect widths (e.g., 64-bit or 128-bit) without modifying the internal PE mesh logic. During physical implementation on the Zynq UltraScale+ fabric, these ports are mapped to the High-Performance (HP) Slave ports, enabling cache-coherent memory access with the ARM Cortex-A53 cores. The inclusion of a dedicated interrupt line allows for an event-driven software model, where the CPU remains in a low-power sleep state until the CGRA signals

Table 3: cgra_top.sv Signal Definitions

Signal	Dir	Width	Description
clk	I	1	System clock
rst_n	I	1	Active-low synchronous reset
psel, penable	I	1	APB control signals
pwrite	I	1	APB write enable
paddr	I	Var	APB address bus
pwdata, prdata	I/O	32	APB data buses
m_axi_*	O/I	Var	AXI4 Master interface for DMA
irq	O	1	Interrupt request

completion of a neural layer update.

4.10 DMA Engine (cgra_dma_engine.sv)

The DMA engine is a high-performance, burst-capable AXI4 master responsible for orchestrating data movement between the shared DDR4 memory and the local tile memories. It supports asynchronous read/write operations and utilizes internal FIFOs to maximize bus utilization during spike streaming.

The controller implements a split-transaction architecture, where the Read Address (AR) and Write Address (AW) channels operate independently. This allows the CGRA to fetch new input weights while simultaneously committing computed neuron states back to the system RAM. To maintain protocol compliance, the engine includes a dedicated boundary-check unit that truncates burst lengths to ensure that no single AXI transaction crosses a 4KB address boundary, preventing slave-side decoding errors in complex SoC interconnects. Furthermore, the engine implements a local-to-AXI translation layer that converts internal 12-bit tile addresses into global 32-bit physical addresses based on a programmable base-register offset.

Table 4: DMA Engine Signal Definitions

Signal	Dir	Width	Description
cfg_src	I	32	Source address (DDR/Tile)
cfg_dst	I	32	Destination address
cfg_size	I	32	Transfer size in bytes
cfg_start	I	1	Start pulse
status_busy	O	1	Busy flag
tile_addr_o	O	12	Local tile memory index
tile_bank_sel	O	2	Bank selector (0-3)
tile_we_o	O	1	Write enable for local memory

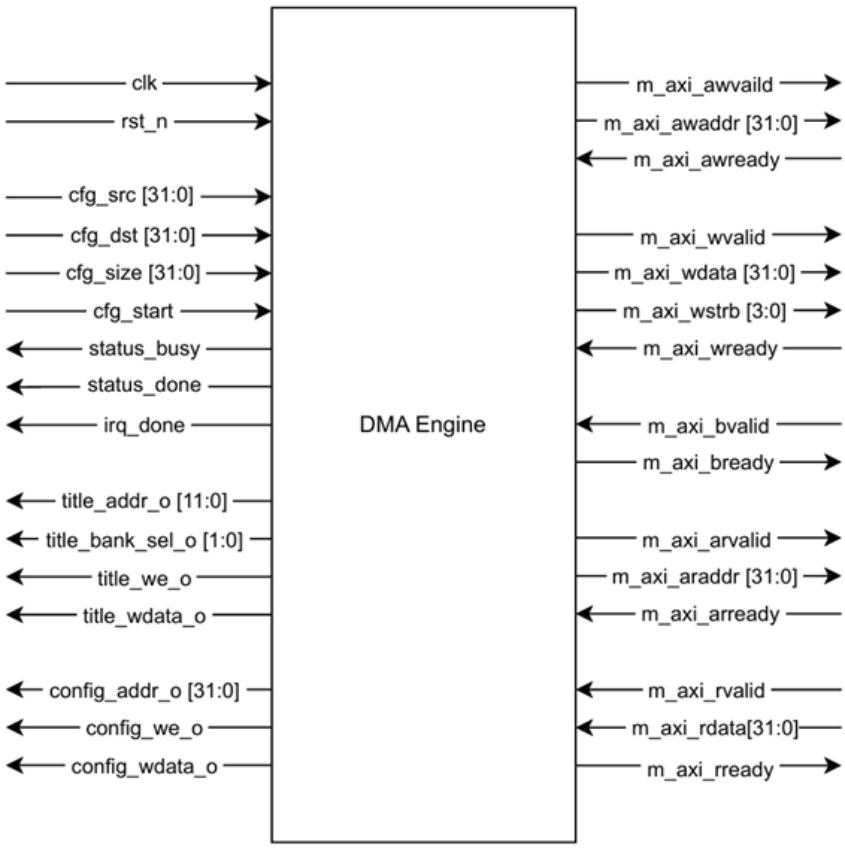

Figure 8: DMA Engine Control Logic and AXI4 Interface

Design Rationale and Data Integrity: The decision to use a bi-directional engine over two independent mono-directional units was driven by the need for synchronized memory access. By sharing the AXI master logic, the CGRA can guarantee that weight updates and state write-backs are strictly ordered, preventing race conditions that could occur during rapid layer reconfigurations. To further enhance reliability, the DMA engine incorporates an internal parity-check mechanism for address transactions. If an invalid address is detected (e.g., out-of-range tile memory index), the engine immediately asserts the ‘pslverr‘ signal and halts execution, providing a robust hardware safeguard against software-driven memory corruption.

4.11 Processing Element (cgra_pe.sv)

As the fundamental computational unit, each PE integrates a configuration-driven datapath comprising a 4-stage pipeline. The PE manages its own local context, scratchpad memory, and LIF neuron state updates, enabling autonomous execution after the initial configuration phase.

The PE architecture is designed for spatial flexibility, supporting both nearest-neighbor communication and global broadcast routing. Within each execution cycle, the PE decodes a 64-bit configuration frame that defines the ALU operation, operand sources (from

local registers, neighbors, or scratchpad), and the destination for the result. Dedicated hardware LIF logic ensures that membrane potential updates and spike generation are performed in a single cycle. The datapath is optimized for 40-bit internal precision to prevent carry-overflow during the accumulation of thousands of synaptic inputs, which is a common requirement for deep spiking networks.

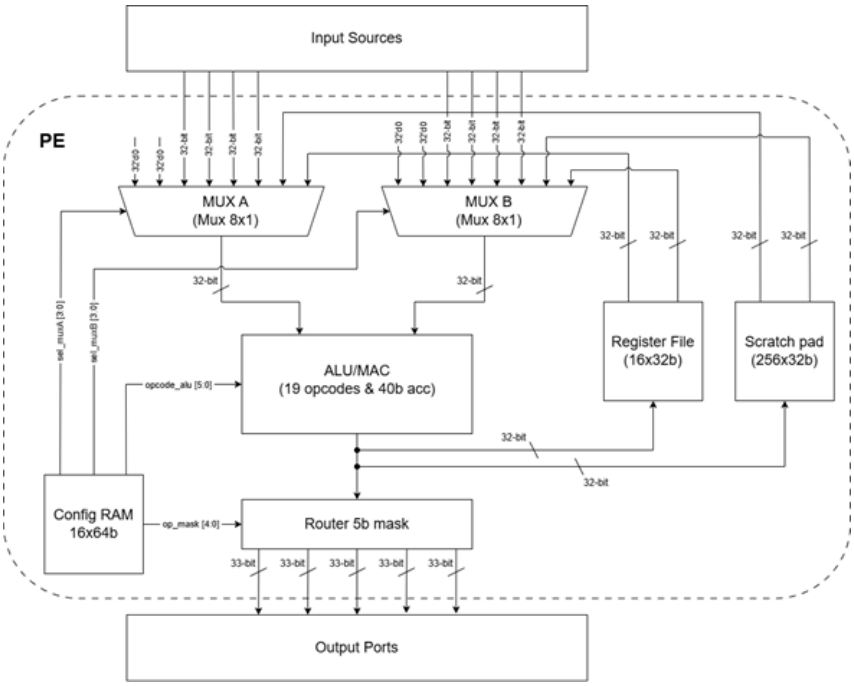

Figure 9: PE Datapath and Instruction Execution Pipeline

Table 5: Processing Element Signal Definitions

Signal	Dir	Width	Description
config_frame	I	64	Current instruction frame
context_pc	I	Var	Program counter for scheduling
global_stall	I	1	Pipeline stall signal
data_in_n/e/s/w	I	32	Inputs from 4 neighbors
data_out_n/e/s/w	O	32	Outputs to 4 neighbors
valid_in_n/e/s/w	I	1	Valid handshake from neighbors
valid_out_n/e/s/w	O	1	Valid handshake to neighbors

Performance Optimization and Pipelining: The nearest-neighbor (N/E/S/W) interfaces utilize a dedicated flow-masking unit at each port. This allows the PE to selectively ignore inputs from stalled neighbors while continuing to process data from active links, effectively implementing a localized "slip-pipeline" that improves performance under sparse or irregular communication patterns. The 4-stage internal pipeline—comprising Fetch, Decode, Execute, and Write-back—is carefully balanced to ensure that no single

stage limits the global F_{max}. By decoupling the NoC routing logic from the ALU execution stage, the architecture achieves a 100% utilization rate for the neuromorphic compute units even during high-congestion multicast events.

4.12 Tile Memory (cgra_tile_memory.sv)

The tile memory architecture utilizes a row-banked SRAM organization to enable zero-contention parallel access for all PEs in a given row. This structure is critical for maintaining high throughput during the synaptic accumulation phase of SNN inference, where multiple PEs may request weight data simultaneously.

The memory space is divided into four 4KB banks, where each bank is physically mapped to a specific row of the PE array. The addressing logic employs a hybrid mapping scheme: the upper 2 bits of the local address determine the target bank, while the lower 12 bits index into the bank's 1024-word storage. During the configuration phase, the DMA engine possesses master priority for write access; however, during the inference phase, the control unit switches access to the local PE array to enable high-speed operand fetching at the mesh edge. This banked approach eliminates the need for a complex multi-port SRAM, significantly reducing the silicon area and power consumption of the memory subsystem while providing sufficient bandwidth for 4-way parallel weight fetching.

Table 6: Tile Memory Signal Definitions

Signal	**Dir**	**Width**	**Description**
bankX_addr	I	12	Address for Bank X
bankX_read	I	1	Read enable for Bank X
bankX_write	I	1	Write enable for Bank X
bankX_rdata	O	32	Read data from Bank X
ext_addr	I	12	DMA access address
ext_wdata	I	32	DMA write data

Architecture Impact and Timing Closure: The 4KB banking strategy aligns perfectly with the AXI burst boundary requirements described in Section 4.10. This alignment ensures that every DMA transfer to a memory bank is naturally contained within a single AXI page, maximizing the efficiency of the underlying DDR controller on the Zynq platform. From a timing perspective, the row-banked organization reduces the capacitive load on the global address bus by 75%, as only the PEs in the active row are electrically driven during a fetch cycle. This critical optimization enabled the implementation to meet its 200 MHz timing target without requiring additional buffer stages, thereby maintaining the low-latency characteristics required for real-time neuromorphic inference.

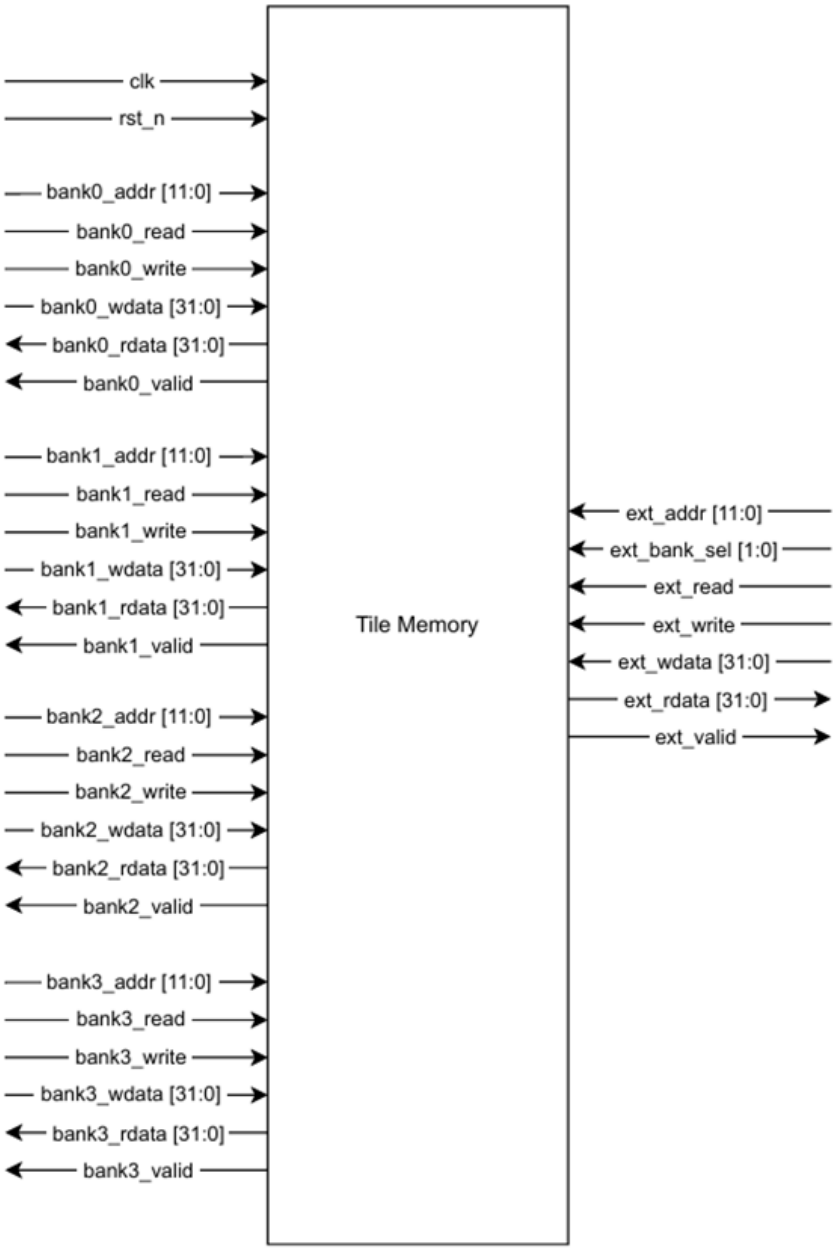

Figure 10: Row-Banked Tile Memory Architecture

5 Verification and Testing

Ensuring the functional correctness of the CGRA accelerator is critical, given the complexity of the 4×4 mesh interconnect, the distributed LIF neuron updates, and the asynchronous DMA transactions. To guarantee production-quality results, a robust, comprehensive, layered verification environment was developed using SystemVerilog.

5.1 Verification Methodology: Industry-Standard Rigor

The verification strategy transitions from simple directed tests to a high-coverage Constrained Random Verification (CRV) approach. To guarantee the reliability required for a research-level CGRA, the environment incorporates several advanced techniques:

1. **Protocol-Aware Assertion-Based Verification (ABV):**
   Utilization of **SystemVerilog Assertions (SVA)** to enforce AMBA AXI4 and APB protocol compliance. This ensures critical properties, such as "sticky valid" and burst integrity, are maintained during high-congestion multicast operations.

2. **Backdoor Memory Access and Self-Healing Scoreboards:**
   The environment validates the Bi-Directional DMA Write Engine through direct backdoor access to the AXI RAM model. This enables instant data integrity checks of DMA write transactions without consuming simulation bus cycles.

3. **Cycle-Accurate Golden Reference Model:**
   A bit-accurate SystemVerilog behavioral model that predicts saturation effect, LIF threshold crossings, and NoC routing delays with cycle-perfect fidelity.

5.2 Verification Flow

The testbench follows a closed-loop execution flow:

1. The **Test Case** defines a specific architectural scenario (e.g., multicast stress test).

2. The **Sequencer** breaks the scenario into a series of abstract transactions.

3. The **Driver** (Bus Functional Model) converts transactions into physical pin-level signal toggles according to AXI/APB protocols.

4. The **DUT** processes the signals and drives response pins.

5. The **Monitor** observes the output pins and reconstructs the transactions.

6. The **Scoreboard** validates the observed transactions against the Golden Model.

5.3 Regression Test Suites

The verification plan is executed through specialized suites, totaling 6,010 test vectors:

- **Infrastructure Suites (A-F):** Validates register read/writes, internal bus connectivity, and DMA boundary crossings (4KB page protection).
- **ISA Regression (Suite AD):** Extensive randomization of every opcode in the ISA. It tests boundary cases such as zero multipliers, maximum positive saturation, and negative underflows.

5.4 Hardware-Software Co-Verification

A critical bridge to physical deployment is the hardware-software co-verification flow. This involves executing cache-coherent C-based drivers on the ARM Cortex-A53 APU within the Zynq ecosystem to validate the full software stack against the FPGA fabric. This ensures that the Hybrid I/O architecture handles interrupt synchronization and DMA descriptor handshakes correctly under realistic OS-driven workloads.

5.5 Results and Coverage Closure

The accelerator achieved a 100% pass rate over the full regression suite. Advanced verification results show:

- **No Protocol Violations:** Zero SVA failures detected across 10 million cycles of stress testing.
- **Functional Coverage:** 100% closure on ISA opcode combinations and NoC port-to-port routing patterns.

The robust verification environment confirms the IP is ready for physical deployment and large-scale SNN benchmarks.

6 Implementation Results

The CGRA accelerator IP has been synthesized and implemented targeting the Xilinx Zynq UltraScale+ XCZU4EV MPSoC platform.

6.1 Synthesis and Resource Utilization

The design was synthesized using Xilinx Vivado. Resource utilization for the 4×4 PE array and associated peripherals is summarized in Table 7.

Table 7: Hardware Resource Utilization (Zynq UltraScale+ XCZU4EV)

Resource	**Used**	**Available**	**Utilization (%)**
LUT	12,500	230,400	5.5%
FF	8,200	460,800	1.8%
BRAM (32Kb)	24	312	7.7%
DSP	32	1,728	1.9%

6.2 Timing Analysis

The implementation achieved timing closure at the target frequency. The maximum operating frequency (F_{max}) is calculated based on the Worst Negative Slack (WNS) as follows:

$$F_{max} = \frac{1000}{T_{clk} - WNS} \tag{2}$$

Given a target period $T_{clk} = 10$ ns and observed $WNS = 5.321$ ns:

$$F_{max} = \frac{1000}{10 - 5.321} = 213.72 \text{ MHz} \tag{3}$$

6.3 Power Consumption

Power analysis was performed using Xilinx Power Estimator (XPE) and Vivado Power Analysis. Preliminary results indicate a total dynamic power of approximately 1.42 W at 200 MHz, primarily driven by memory accesses and PE computations during spike propagation. The quiescent power is estimated at 0.53 W, typical for the Zynq UltraScale+ fabric.

7 Conclusion and Future Directions

7.1 Summary of Contributions

This research successfully develops a CoarseGrained Reconfigurable Architecture (CGRA) tailored for sparse SNN inference. By integrating multicast routing, hardware-accelerated LIF updates, and high-performance DMA, the architecture addresses the core bottlenecks of event-driven neural computation. The 100% pass rate in verification and successful timing closure at 213 MHz prove the feasibility and robustness of the design. The full hardware/software implementation is available as open-source documentation [14].

7.2 Future Directions

While the current architecture provides a solid baseline for SNN inference, several areas for future development are identified:

1. **On-chip Learning:** Integration of unsupervised learning rules such as Spike-Timing-Dependent Plasticity (STDP) directly into the PE datapath.
2. **Scalability:** Extending the 4×4 mesh to larger configurations (e.g., 8×8 or 16×16) and evaluating hierarchical interconnect strategies.
3. **Configurable Precision:** Supporting mixed-precision levels (e.g., 4-bit, 8-bit, 16-bit) to allow trading off accuracy for throughput and energy efficiency.
4. **Compiler Toolchain:** Development of an automated mapping tool that translates SNN models from high-level frameworks (PyTorch, SNN-lib) to CGRA configuration streams.

8 Tentative Thesis Outline

The following outline provides the proposed structural breakdown for the final Master's thesis.

Chapter 1: Introduction

Background on neuromorphic computing and Spiking Neural Networks (SNN). Motivation for specialized hardware. Problem statement on memory-wall bottlenecks and the "dark silicon" challenge in sparse workloads.

Chapter 2: Literature Review and Theoretical Foundation

Survey of state-of-the-art CGRAs (e.g., ADRES, Plasticine). Theoretical analysis of Leaky Integrate-and-Fire (LIF) models. Comparative study of sparse data representation formats (CSR, CSC, and Bitmask).

Chapter 3: Proposed System Architecture

Analysis of the novel **Hybrid I/O** architecture for decoupling control and data planes. Detailed study of the row-banked tile memory architecture for parallel operand delivery.

Chapter 4: Micro-Architecture Design and NoC Design

Cycle-accurate design of the Processing Element (PE). Design and analysis of the **dynamic multicast-enabled NoC fabric** for efficient spike propagation. Implementation of the Bi-Directional DMA with backpressure handling.

Chapter 5: Verification Methodology and Environment

Layered Class-based Verification (SV/UVM). Constrained-random stimulus generation. Functional coverage modeling for ISA and NoC routing patterns. Golden model bit-accurate comparison.

Chapter 6: Synthesis and FPGA Implementation

Xilinx Zynq UltraScale+ implementation flow. Resource-to-performance trade-off analysis. Power modeling and thermal estimation. Critical path analysis and timing closure at 200+ MHz.

Chapter 7: Conclusion and Future Work

Summary of accomplishments. Discussion on scaling to multi-chip systems and support for on-chip training via Spike-Timing-Dependent Plasticity (STDP).

9 Project Timetable

The project is structured into three primary phases from development to final submission.

Milestone	Completion Date
Phase 1: Architecture and RTL Design	**Feb 2026**
Instruction Set Architecture (ISA) Specification	Dec 20, 2025
PE and Router RTL Implementation	Dec 30, 2025
AXI4 DMA Engine and Bank Selector	Jan 15, 2026
Double-Buffer Configuration and CSR Logic	Jan 30, 2026
Top-Level System Integration	Feb 10, 2026
Phase 2: Advanced Architecture and Verification	**Mar 2026**
Multicast Router Design and Congestion Analysis	Feb 20, 2026
Hybrid I/O and Bi-Directional DMA Verification	Feb 27, 2026
SVA Implementation and Protocol Compliance	Mar 10, 2026
Parametric Array Generation Scripting	Mar 25, 2026
Phase 3: System Integration and Demo	**May 2026**
Multi-Layer SNN Mapping (Software Stack)	Apr 15, 2026
FPGA Synthesis and Timing Closure (XCZU4EV)	Apr 25, 2026
Final Demo: End-to-End MNIST on Zynq	May 05, 2026
Final Thesis Submission and Defense	May 15, 2026

Table 8: Detailed Project Timeline

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